

ALG-DATA-01 — The Mystery Dataset

1. Project Title

The Mystery Dataset — Discovering Hidden Patterns in Online Retail Transactions

2. Problem Statement

The objective of this project is to investigate a large dataset without a predefined business question and discover meaningful patterns, relationships, anomalies, and insights through systematic data analysis.

The project uses the **Online Retail Dataset** from the UCI Machine Learning Repository.

3. Dataset

The dataset contains transactions from an online retail business.

Dataset characteristics

- **Rows:** 541,909
- **Columns:** 8
- **Time period:** December 2010 – December 2011
- **Countries:** Multiple countries
- **Main variables:**
 - InvoiceNo
 - StockCode
 - Description
 - Quantity
 - InvoiceDate
 - UnitPrice
 - CustomerID
 - Country

An additional derived variable, TotalAmount, was calculated as:

TotalAmount = Quantity × UnitPrice

4. Technologies Used

- Python
 - Pandas
 - Matplotlib
 - Excel dataset
 - Exploratory Data Analysis (EDA)
-

5. Analysis Methodology

The analysis followed a discovery-driven approach rather than starting with a fixed business question.

Step 1 — Dataset Understanding

The dataset structure, variables, data types, statistical characteristics, and missing values were examined.

Step 2 — Data Quality Investigation

Missing values and unusual values were investigated, especially:

- Missing CustomerID
- Missing Description
- Negative Quantity
- Zero UnitPrice
- Unusual invoice numbers

Step 3 — Anomaly Detection

Transactions with negative quantities were investigated.

A total of:

10,624 transactions contained negative quantities.

Among these:

- **9,288** had a C-prefixed invoice number.
- **1,336** did not have a C-prefixed invoice number.

This showed that negative quantity transactions were not a single uniform category.

Step 4 — Deeper Investigation

The 1,336 negative-quantity transactions without a C invoice prefix were investigated further.

A notable pattern was discovered:

- Negative quantity
- Zero UnitPrice
- Strong concentration in the United Kingdom
- Descriptions containing terms such as:
 - damaged
 - damage
 - check
 - destroyed
 - thrown away
 - missing
 - wet
 - adjustment
 - unsaleable
 - smashed

Step 5 — Relationship Analysis

Relationships between transaction characteristics were examined, including:

- Quantity and UnitPrice
- Invoice type and negative quantities
- Country and missing CustomerID

- Country and zero-price negative transactions
- Product and negative quantity
- Time and unusual transactions

Step 6 — Visualization

Multiple visualizations were created to make the discovered patterns easier to interpret.

The visualizations include:

1. Negative Quantity Transactions by Invoice Type
 2. Top Descriptions in Zero-Price Negative Transactions
 3. Products with Highest Negative Quantity
 4. Monthly Pattern of Zero-Price Negative Transactions
-

6. Major Discovery

The strongest discovery from the investigation was the existence of a distinct group of transactions that differs from ordinary sales and from the majority of cancellation-style transactions.

These transactions commonly contain:

- Negative quantity
- Zero unit price
- No C-prefixed invoice
- UK concentration
- Operational descriptions related to damage, checking, disposal, missing items, or adjustments

This suggests that the dataset contains **operational or inventory-related activity mixed with normal customer transactions**.

The analysis does not claim that every such transaction represents a specific operational event. Instead, the combination of multiple independent signals provides evidence for a distinct transaction pattern.

7. Hypothesis

Hypothesis

A subset of negative-quantity, zero-price transactions may represent operational inventory adjustments or non-sales activities rather than ordinary customer cancellations.

Evidence

The hypothesis is supported by:

- 1,336 negative-quantity transactions without a C invoice prefix.
 - These transactions have zero UnitPrice.
 - They are concentrated in the United Kingdom.
 - Their descriptions frequently contain operational terms associated with damage, checking, disposal, missing items, or adjustments.
-

8. Analytical Result

The investigation demonstrates that simply classifying every negative quantity as a cancellation would overlook a second meaningful pattern.

Therefore, invoice prefix, quantity, price, description, country, and transaction date should be considered together when interpreting unusual transactions.

This demonstrates the importance of **multi-variable exploratory analysis** in an unknown dataset.

9. Conclusion

The Online Retail dataset initially appeared to be a conventional collection of sales transactions. However, systematic investigation revealed hidden transaction patterns that were not obvious from the dataset structure alone.

The strongest discovery was a distinct group of zero-price negative-quantity transactions with operational descriptions and strong UK concentration.

The project demonstrates how exploratory data analysis can transform an undefined dataset into an evidence-backed analytical story.

Final Insight

The most important finding is not simply that negative transactions exist, but that negative transactions themselves contain different behavioral patterns that can be separated using invoice type, price, description, country, and quantity.

10. Future Scope

The discovered pattern could be investigated further by:

- Developing an automated anomaly classification system
 - Applying NLP to transaction descriptions
 - Creating an interactive analytics dashboard
 - Comparing operational anomalies across countries
 - Building a predictive model for unusual transactions
 - Investigating product-level anomaly behavior over time
-

11. Project Outcome

The project successfully demonstrates:

- Data exploration
- Data quality investigation
- Variable understanding
- Relationship analysis
- Anomaly detection
- Visualization
- Hypothesis formation
- Evidence-backed analytical discovery